

7 · The tournament: scikit-learn on real data

Slide 90 · P&M 7.4

i The question: four learners, one data set. Which one spots counterfeit banknotes best, and how sure can we be of the ranking?

The data (banknotes.csv, in the course folder *Counterfeit banknotes with scikit learn*): 1372 banknotes. Four numbers per note, measured from a photo of it (variance, skewness, curtosis, entropy), and a label: 0 authentic, 1 counterfeit.

Every scikit-learn model works the same way: make it, fit it on the training set, predict on the test set, and compare with the true labels. `train_test_split(..., test_size=0.4, random_state=seed)` is the holdout split from worksheet Task 9: 60% training, 40% test. The `random_state` (the *seed*) decides which notes land in which set.

The four players: Perceptron (Task 4), SVC (a support vector machine, Task 6), `KNeighborsClassifier(n_neighbors=1)` and `(n_neighbors=3)` (Task 3).

Task 12 · Round 1: all four features. Work in pairs; swap who types after M2.

Start a file `tournament.py` in the folder that holds `banknotes.csv`, or add a cell to `Learning_exercise.ipynb`.

```
import csv
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier

# M1: read the data (given)
with open("banknotes.csv") as f:
    reader = csv.reader(f)
    next(reader) # skip the header row
    evidence, labels = [], []
    for row in reader:
        evidence.append([float(cell) for cell in row[:4]])
        labels.append(int(row[4])) # 1 = counterfeit

def accuracy(model, X, y):
    """Share of correct predictions: 1 minus the average 0-1 loss."""
    predictions = model.predict(X)
    return sum(p == t for p, t in zip(predictions, y)) / len(y)

# M2: holdout split, seed 0
X_train, X_test, y_train, y_test = train_test_split(
    evidence, labels, test_size=___, random_state=0)

# M3: one model
model = Perceptron(random_state=0)
model.___(X_train, y_train) # learn from the training set only
print(f"train {accuracy(model, ___, ___):.3f} test {accuracy(model, ___, ___):.3f}")
```

Code ladder. Tick a step only when you can show it works.

	Milestone	Check
M0	import sklearn runs, and banknotes.csv is in the folder	no error
M1	data read	len(evidence) is 1372; sum(labels) is 610
M2	split done	823 training notes, 549 test notes
M3	one Perceptron, seed 0	train 0.984, test 0.980
M4	all four models, seeds 0–4, written on the board	20 rows, each with train and test

M4: the loop. Fill in the blanks, run, and copy the *mean test accuracy* of each model into the table and onto the board.

```
models = {
    "Perceptron": lambda seed: Perceptron(random_state=seed),
    "SVC": lambda seed: SVC(),
    "1-NN": lambda seed: KNeighborsClassifier(n_neighbors=1),
    "3-NN": lambda seed: KNeighborsClassifier(n_neighbors=___),
}
for name, make in models.items():
    tests = []
    for seed in range(5):
        X_train, X_test, y_train, y_test = train_test_split(
            evidence, labels, test_size=0.4, random_state=___)
        model = make(seed)
        model.fit(X_train, y_train)
        tests.append(accuracy(model, X_test, y_test))
    print(name, seed, f"{accuracy(model, X_train, y_train):.3f}", f"{tests[-1]:.3f}")
print(name, "mean test", f"{sum(tests) / len(tests):.3f}",
      "min/max", f"{min(tests):.3f}", f"{max(tests):.3f}")
```

model	mean test accuracy	lowest test accuracy	highest test accuracy
Perceptron			
SVC			
1-NN			
3-NN			

Some models differ by less than 0.01 in mean test accuracy. Is that a real difference? Look at how far one model moves between seeds. _____

Done when: M0–M4 ticked, the table filled, and your answer about the ranking.

Task 13 · Extension if time: Round 2, only two features

Keep only the first two numbers of each note (variance and skewness): add `two = [row[:2] for row in evidence]` and use `two` instead of `evidence` in the loop.

Predict first: for 1-NN: training accuracy _____ test accuracy: higher, equal or lower? _____

Why do you expect that training accuracy? (worksheet Task 9) _____

model	mean training accuracy	mean test accuracy	gap (train – test)
Perceptron			
SVC			
1-NN			
3-NN			

For the mean training accuracy, collect the training numbers the loop prints (or add a second list next to tests).

(1) Which model has the largest gap? What is that gap evidence of? _____

(2) With two features, the perceptron's test score swings more between seeds than the others'. Suggest one reason. _____

(3) The club in the worksheet had 15 days; here there are 1372 notes. Why do the seeds matter less here than the four holdout days did in Task 9? _____

Done when: if doing this extension: a prediction before running, the table and three answers.

Optional extension · Round 3: compare k by cross-validation

```
from sklearn.model_selection import cross_val_score
for k in (1, 3, 5, 15, 51, 151):
    scores = cross_val_score(KNeighborsClassifier(n_neighbors=k), two, labels, cv=5)
    print(k, f"{scores.mean():.3f}")
```

Which k has the highest validation score? _____

This is exploratory: selecting a model using these scores needs a new untouched test for a final performance claim.

Optional after-class consolidation: `learning_by_hand.py`. Choose steps to practise; it is ungraded and not a prerequisite for the final assignment.

Notes